

4.4.1 WAP

Wireless Application Protocol (WAP) is an open international standard for applications that use wireless communication, for example, Internet access from a mobile phone. WAP was designed to provide services equivalent to a Web browser with some mobile-specific additions, being specifically designed to address the limitations of very small portable devices. It's now the protocol used for the majority of the world's mobile Internet sites, otherwise known as WAP-sites. The Japanese i-mode system is the other major competing wireless data protocol.

WAP is a protocol suite created for mobile devices such as PDAs and mobile phones, aiming at giving their users a richer data application experience that would enable "access to the Internet," as it was then put. Before WAP became popular, it was nearly impossible for application providers to build interactive data applications that would allow, for example, business people on the go to catch an important e-mail or learn early on that they were losing money on their stocks; neither was it possible to build rich data applications allowing consumers to get sports results and the latest news headlines from their TV. WAP aimed at enabling this type of applications to be built on wireless technology.

Before the WAP protocol was created, wireless handset data application capabilities were limited to rudimentary interfaces such as SMS or USSD, a rudimentary browsing service. Some manufacturers had also come up with their own, proprietary "enhanced" application platforms such as HDML (by Unwired Planet, now Openwave), ITTP by Ericsson and TTML by Nokia. However, to enhance the overall size of their business opportunity, manufacturers decided to join forces on technology and extended an invitation to the rest of the wireless industry in building a common standard for wireless data applications.

The WAP Forum proposed a protocol suite that would allow the interoperability of WAP equipment and software with many different network technologies; the rationale for this was to build